Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21321

F-421

B.C.A. (Fifth Semester) Examination, 2023-24

(New Course)

Paper - BCA-501-N

Computer Graphics And Animation

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What do you mean by computer graphics? Explain its role in education.

   (b) Explain various types of video file formats.

2. (a) Describe color display technique.

   (b) How number of pixels in any image effects its quality? Justify clearly.

3. (a) What is the concept of anti aliasing?

   (b) Differentiate 2-D viewing concept from 3-D viewing concepts.

4. (a) How to draw a line? Explain any one algorithm.

   (b) Draw and explain the concept of display technology (any one of your choice).

5 (a) What are spline curves? Explain its concept in detail.

   (b) Illustrate the procedure of polygon filling.

6. (a) How to create any animation? Write down the steps.

(b) How you seen the practical application of "Morphing" principle. Describe its working in detail.

7. (a) How impact and non-impact printers differ? Explain.

(b) Differentiate between window port and view port.

8. (a) Write down the steps involved in 3D transformation?

OR

Define active and passive transformation.

(b) Distinguish between convex and concave polygons.

OR

Write down the applications of Animation.

Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21321

F-421

B.C.A. (Fifth Semester)

Examination, 2022-23

(New Course)

Paper-BCA-501-N

COMPUTER GRAPHICS AND ANIMATION

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What is development of computer graphics? Explain it in details. 7

   (b) Describe in details color display techniques. 7

2. (a) Explain the frame buffer and bit operations concepts in raster graphics. 7

   (b) Differentiate between : 7

   (i) Storags displays and display processors

   (ii) Raster scan and Random scan graphics.

3. (a) What is line drawing algorithms? Explain it with an example. 7

   (b) What is conic section generation? Describe it in details. 7

4. (a) Differentiate between ellipse generation and polygon filling. 7

   (b) What is two-dimensional viewing? Explain it with an example. 7

5. (a) Explain in details interactive picture construction techniques. 7

(b) What is 2-D and 3-D transformation? Explain it with an example. 7

6. (a) What are Bezier curves? State the mathematical expression of Bezier curve. 7

(b) What is 3-D viewing? Explain it in details. 7

7. (a) Describe the algorithms for 3-D volumes spline curves and Surfaces in details. 7

(b) How many types of animation? Explain it is details. 7

8. Explain the following : 14

(i) GKS primitive

(ii) Multimedia application

(iii) Morphing

(iv) Tweaking

Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21321

F-421

B.C.A. (Fifth Semester)

Examination, March-2022

(New Course)

Paper - BCA-501-N

Computer Graphics And Animation

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are various elements of computer graphics?

   (b) Discuss different image formats.

2. (a) How raster scan is different from random scan?

   (b) Describe different display technologies.

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3. (a) Explain any line drawing algorithm.

   (b) What is antialiasing?

4. (a) Explain various picture construction techniques.

   (b) Write in brief about the following.

   (i) Clipping

   (ii) 3–D transformation

5. (a) How to draw circle and ellipse? Explain with proper example.

   (b) Differentiate 2–D transformation from 3–D transformation.

6. (a) How Animation is implemented in computer graphics?

   (b) Elaborate the 3–D viewing algorithm.

7. (a) What do you mean by the term "MORPHING"? Write down its practical application giving one suitable example.

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(b) Find the new coordinate of triangle A(0, 0), B(1, 1), C(5, 2) after it has been magnified to twice of its size.

8. (a) Define translation, rotation and scaling.

(b) Define Polygon clipping.

(c) Define recursively defined B–Splines.

(d) Define scalars, point and vectors.

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Paper Code : 21321

F-421

B.C.A. (Fifth Semester) Examination, 2021

(New Course)

Paper-BCA-501-N

COMPUTER GRAPHICS AND ANIMATION

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What is computer Graphics? Write the important applications of Computer Graphics? 7

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(b) Define Random and Raster scan? 7

2. (a) Explain color display techniques, with diagram. 7

(b) Differentiate between: 7

(i) Image and graphics

(ii) Video and animation

3. (a) What is Frame Buffer? Discuss the role of frame buffer in controlling intensity and color of display devices? 7

(b) Explain the limitation of DDA algorithm for line generation. Expand the abbreviation DDA. 7

4. What is 2D Transformation? Explain types of 2D transformations. 14

5. (a) What is Aliasing? Define Anti-aliasing techniques. 7

(b) What are Bezier curves? State the mathematical expression of Bezier curve, explaining the terms involved. 7

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6. What is animation? What are two types of animation? Explain both. 14

7. A homogenous co-ordinate point P(3,2,1) is translated in x, y and z directions by-2, -2 and -2 unit respectively followed by successive rotation 60 degree about x-axis. Find the final position of homogenous co-ordinate. 14

8. Explain the following: 14

(i) Tweaking

(ii) Morphing

(iii) Clipping

(iv) Mid-point circle drawing algorithm

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Paper Code : 21321

F–421

B.C.A. (Vth Semester) Examination, 2019–20

(New Course)

COMPUTER GRAPHICS AND ANIMATION

Paper-BCA-501-N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are the elements of computer graphics ?

   (b) How Raster-Scan is different from Random-Scan ?

2. (a) Describe various types of color display techniques.

   (b) Write down any line drawing algorithm.

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3. (a) What do you mean by 'anti-aliasing' ? Define aspect ratio.

   (b) Explain what is viewing transformation ?

4. What is projection ? Describe various types of projections in detail.

5. Prove that *two* scaling transformations is commutative in nature i.e., $S_1S_2 = S_2S_1$.

6. How animation is different from image ? Explain *different* steps in animation sequences.

7. Write in detail about the following :

   (a) Tweeking

   (b) Morphing

8. (a) What is scan conversion ? Explain.

   (b) Describe various applications of multimedia.

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Paper Code : 21321

F–421

B.C.A. (Fifth Semester) Examination, 2018

(New Course)

Paper-BCA-501-N

COMPUTER GRAPHICS AND ANIMATION

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What do you understand by term computer graphics ? What is the need of it ?
   (b) Explain the basic functionality of Random scan. Display with figure.
2. (a) Explain color display techniques in detail.
   (b) Explain bit operations concepts in raster graphics.
3. (a) Draw a line from (0, 0) to (8, 8) using DDA Algorithms.
   (b) Explain the basic transformation techniques in detail for 2D.

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4. (a) What are the difference between windowing and clipping ?
   (b) What do you understand by term segment ? Explain its structure.
5. (a) Explain transformation technique for 3D images.
   (b) Write down the algorithm for 3-D volumes spline curves.
6. (a) Explain interactive picture construction techniques in detail.
   (b) Explain about interactive input and output devices in detail.
7. (a) Explain GKS primitive in detail
   (b) Explain multimedia applications in detail.
8. Explain any *three* terms :
   (a) Tweaking
   (b) Morphing
   (c) RGB colour model
   (d) Types of animation system.

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B.C.A. (Fifth Semester) Examination, 2018

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Paper-BCA-501-N

COMPUTER GRAPHICS AND ANIMATION

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

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   (b) Explain the basic functionality of Random scan. Display with figure.

2. (a) Explain color display techniques in detail.

   (b) Explain bit operations concepts in raster graphics.

3. (a) Draw a line from (0, 0) to (8, 8) using DDA Algorithms.

   (b) Explain the basic transformation techniques in detail for 2D.

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   (b) Explain multimedia applications in detail.

8. Explain any *three* terms :

   (a) Tweaking

   (b) Morphing

   (c) RGB colour model

   (d) Types of animation system.

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